

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech.

Branch: Mechanical & Allied Engg. Semester: III

Subject Code & Name: BTMES304 Materials Science and Metallurgy

Max Marks: 60

Date: 09-01-24

Duration: 3Hrs.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Sketch the following directions and planes within a cubic unit cell: $[3\bar{1}2]$ $[10\bar{2}]$ $(2\bar{1}1)$ $(3\bar{1}2)$		6
B) Explain, in brief, the slip and twinning mechanisms of plastic deformation with the help of schematic diagrams.		6
C) Compare Brinell Hardness Test with Rockwell Hardness Test.		6
Q.2 Solve Any Two of the following.		12
A) Explain the Hume-Rothery's rules of solid solubility. Show the solidification of pure metals graphically. https://www.batuonline.com		6
B) What is lever rule? Derive the statement of Lever rule using schematic diagrams.		6
C) Discuss the eutectoid transformation on the equilibrium diagram of steels. Draw the microstructure of the eutectoid steel.		6
Q. 3 Solve Any Two of the following.		12
A) Explain the mechanism of removal of heat during cooling with the help of schematic diagrams/graphs. Give the requirements of an ideal quenchant.		6
B) Differentiate clearly between annealing and normalizing heat treatments. Show the temperature range of these treatments on the equilibrium diagram.		6
C) Describe the different types of flame hardening processes. What is the principle of induction hardening treatment?		6

Q.4 Solve Any Two of the following.

- A) Compare macroscopy with microscopy w.r.t. definition, purposes, specimen preparation, merits, and demerits.
- B) Define etching. Differentiate clearly between the mechanism of etching for single phase and two phase alloys.
- C) Explain sulphur print test w.r.t. purpose, procedure, and chemical reactions involved.

Q. 5 Solve Any Two of the following.

- A) Explain dispersion strengthening w.r.t. basic mechanism, critical factors, advantage, and commercial examples
- B) Discuss age/precipitation hardening process w.r.t. basic principle, conditions, and steps.
- C) Describe magnetic particle inspection method using schematic diagrams.

***** End *****

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